SUVEKSHYA JHA

Data Analyst & ML Engineer | AI/NLP Specialist

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PROFESSIONAL SUMMARY

Computer Science undergraduate (BSc. CSIT) focused on Machine Learning, Deep Learning, and Natural Language Processing (NLP), working toward building production-grade AI systems that address real-world problems. Has independently delivered end-to-end projects spanning classical ensemble learning and feature engineering pipelines to Retrieval-Augmented Generation (RAG) systems built with LLaMA 3.x and vector databases, containerized with Docker for reproducible deployment. Comfortable across the full ML lifecycle, including data engineering, exploratory data analysis, feature engineering, LLM fine-tuning, hyperparameter tuning, cross-validation, model optimization, evaluation, and deployment, along with bias-variance tradeoff analysis. Particularly interested in applying transformer architectures and NLP to high-impact domains such as medical AI and misinformation detection.

EDUCATION

BSc. CSIT (Computer Science & Information Technology) — *Asian College of Higher Studies, Ekantakuna, Lalitpur*
Expected: March 2028

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Artificial Intelligence, Statistics & Probability, Software Engineering

+2 Science (Physics, Chemistry, Mathematics, Computer Science) — *Xavier Int'l College, Kalopul, Kathmandu*
Completed: January 2023

TECHNICAL SKILLS

Languages Python (Advanced) · SQL (Intermediate)

Deep Learning Neural Networks · Transformers · Fine-tuning LLMs · LLaMA 3.2/3.3 (HuggingFace Transformers · llama.cpp) · Mistral · Sentence Transformers · BAAI/bge-base-en-v1.5

NLP NLTK · TextBlob · TF-IDF Vectorisation · N-gram Analysis · Text Preprocessing · Sentiment Analysis · Semantic Search · Named Entity Recognition

ML Frameworks Scikit-learn · XGBoost · LightGBM · Optuna · RandomizedSearchCV · Sklearn Pipelines

ML Algorithms Logistic Regression · Random Forest · Gradient Boosting · Decision Tree · KNN · SVM · Ensemble Learning · Cross-Validation · Hyperparameter Tuning · Bias-Variance Tradeoff

Model Skills Model Deployment · Model Optimization · Evaluation Metrics (F1, ROC-AUC, Precision, Recall, AUC) · Model Benchmarking

Data Engineering Feature Engineering · Data Cleaning · EDA · Statistical Analysis · Pandas · NumPy

Data Visualization Matplotlib · Seaborn · Plotly · WordCloud · TF-IDF N-gram Charts

LLMs & RAG Retrieval-Augmented Generation (RAG) · Prompt Engineering · LangChain · ChromaDB · Semantic Similarity Search · Embedding Indexing · Dual Vector Store Design · Ollama (local inference)

Deployment & Infra Docker · FastAPI · Streamlit · REST APIs · ngrok · HTML/CSS Frontend

Tools Git · GitHub · Jupyter Notebook · VS Code · python-dotenv · Groq API

WORK EXPERIENCE

Data Science & AI/ML Training + Internship — *DLytica Inc.* **Nov 2025 – Mar 2026**

Remote · 4-month professional training and internship program

- Completed a structured 4-month Data Science with AI/ML curriculum covering data engineering, exploratory data analysis, statistical analysis, feature engineering, ML algorithms, model evaluation, and NLP/AI application development, reinforcing the technical foundation applied across independent projects.
- Delivered a capstone project spanning end-to-end EDA and data cleaning, supervised ML model development and evaluation, and an NLP/AI-based application component, applying the same model evaluation metrics and feature engineering practices used in personal projects.
- Received a Certificate of Professional Training with Internship (ISO 27001:2022 certified program) in recognition of consistent performance, dedication, and professionalism throughout the program.

PROJECT EXPERIENCE

Medical Report Q&A System — **Jan 2026 – Apr 2026**

Challenge: Patients and clinicians struggled to locate specific information within dense, multi-page medical reports, making manual review slow and error-prone. Designed and delivered a production-grade RAG system that answers plain-language questions against uploaded PDFs with cited, context-grounded responses.

- Architected a dual-vector-store system using ChromaDB and BAAI/bge-base-en-v1.5 Sentence Transformer embeddings to manage 50,000+ data chunks from 4,000 medical transcripts. Adopted the dual-store design after single-store retrieval produced inconsistent ranking on long documents; the change reduced response latency by 30% and improved answer accuracy by 20% based on user feedback.
- Engineered a priority-cascade retrieval strategy with model optimization tuning, testing retrieval depths from top-3 to top-10 chunks and selecting the configuration that improved answer relevance by 25%, cut irrelevant responses by 15%, and increased retrieval efficiency by 20% over the baseline single-pass approach.
- Integrated LLaMA 3.3-70B via the Groq API with a strict medical prompt template enforcing source citation per claim, urgent-item flagging, and jargon simplification, directly reducing the risk of medical misinformation in a high-stakes domain.
- Applied an L2-distance-to-similarity conversion (score = 1/(1+distance)) for interpretable chunk scoring, and built a Plotly relevance bar chart with green/amber/red thresholds to make model confidence transparent to non-technical clinicians.
- Containerized the FastAPI backend with Docker for reproducible deployment, and benchmarked Mistral against LLaMA 3.3-70B on response latency and answer quality to select the better-performing model for production use.
- Designed idempotent CSV indexing to skip re-embedding on subsequent starts, reducing cold-start overhead to near zero, and structured each module (utils.py, rag_pipeline.py, llm_answer.py) for isolated testing and future RAGAS faithfulness evaluation.
- Delivered a multi-session Streamlit dashboard with streaming word-by-word output, session history, and expandable source panels, communicating complex AI outputs clearly to end users with no ML background.

Tech Stack: Python · LangChain · LLaMA 3.3-70B (Groq API) · Mistral · ChromaDB · BAAI/bge-base-en-v1.5 · Sentence Transformers · PyMuPDF · Streamlit · Plotly · Pandas · NumPy · Docker

PDF-Based RAG Chatbot System — **Jan 2026**

Challenge: Professionals and students faced the tedious task of reading entire PDFs, such as research papers, manuals, and legal documents, to find specific answers. Built a local, zero-cloud-cost chatbot that turns any static PDF into an interactive knowledge source.

- Developed a PDF ingestion pipeline using NLP and semantic embeddings with Sentence Transformers (all-MiniLM-L6-v2), improving answer completeness by 40% and reducing manual document processing time by 50 hours annually in pilot testing.
- Stored dense vector embeddings in ChromaDB to enable semantic search that correctly handles paraphrase and synonym queries, validating retrieval quality via semantic similarity scores rather than keyword overlap alone.
- Integrated Meta's LLaMA 3.2-1B-Instruct as the local LLM, grounded strictly in retrieved document chunks, applying bias-variance tradeoff reasoning to select a compact model that avoids hallucination on domain-specific technical content.
- Also experimented with Ollama for local LLM serving and inference, complementing production deployments via Groq API and llama.cpp.
- Deployed via a FastAPI backend with a custom HTML/CSS frontend, tunnelled through ngrok for live demonstration, achieving full model deployment without any cloud infrastructure or hosting costs.

Tech Stack: Python · LLaMA 3.2-1B (llama.cpp / HuggingFace Transformers) · Sentence Transformers · ChromaDB · FastAPI · PyPDF · ngrok · HTML/CSS · Ollama

Fake News Detection using NLP and Ensemble Learning — **Nov 2025**

Challenge: Manual fact-checking does not scale, and misinformation spreads faster than human reviewers can respond. Designed and validated an automated NLP and ensemble-learning pipeline to act as a scalable first-pass filter for editorial workflows.

- Conducted exploratory data analysis on 5,000 news articles using data visualization techniques (WordClouds, TF-IDF n-gram charts, pairplots), identifying linguistic patterns that distinguished fake from real articles with 85% accuracy and informing all downstream feature choices.
- Engineered 5,000 TF-IDF bigram features with regex-based text preprocessing, polarity and subjectivity scores from TextBlob, and a combined title-and-body representation, yielding a 10% increase in precision and recall over single-field approaches, validated through cross-validation.
- Applied ensemble learning (hard and soft voting across Logistic Regression, Random Forest, and XGBoost) with hyperparameter tuning via Optuna and RandomizedSearchCV, selecting the final configuration by minimizing the bias-variance tradeoff and achieving F1 = 0.88 and AUC = 0.95 on held-out test data.
- Achieved 92% overall accuracy with a 40% reduction in false positives during pilot deployment; the resulting model optimization cut manual review workload by approximately 15 hours per week, demonstrating viable production use as an editorial pre-filter.
- Made model decisions interpretable to non-technical stakeholders through top-20 word-frequency bar charts, WordClouds, and n-gram visualizations, reinforcing technical communication skills critical for cross-functional teams.

Tech Stack: Python · Pandas · NumPy · Scikit-learn · NLTK · TextBlob · WordCloud · TF-IDF Vectorizer · Matplotlib · Seaborn · XGBoost · Optuna · RandomizedSearchCV

SOFT SKILLS & LEADERSHIP QUALITIES

- Problem Decomposition & First-Principles Thinking
- Self-Directed Learning & Initiative
- Technical Communication & Documentation
- End-to-End Project Ownership
- Analytical & Data-Driven Decision Making
- Research Aptitude & Applied Learning
- Adaptability & Domain Curiosity

LANGUAGES

English — Advanced · Nepali — Native · Hindi — Conversational